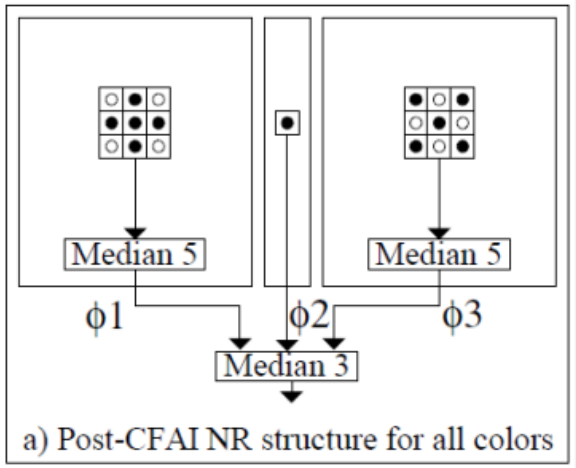
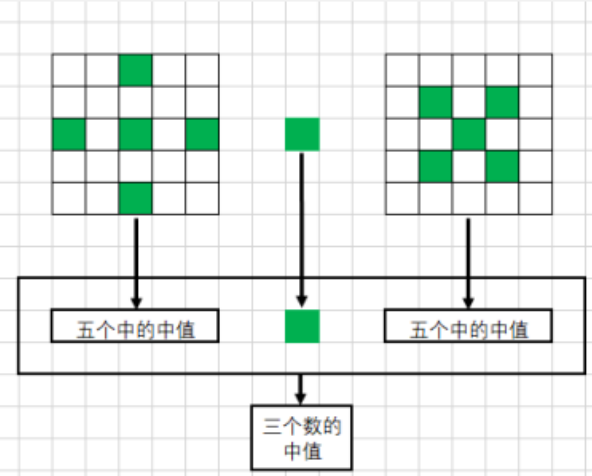
中值滤波家族

参考：B站食鱼者

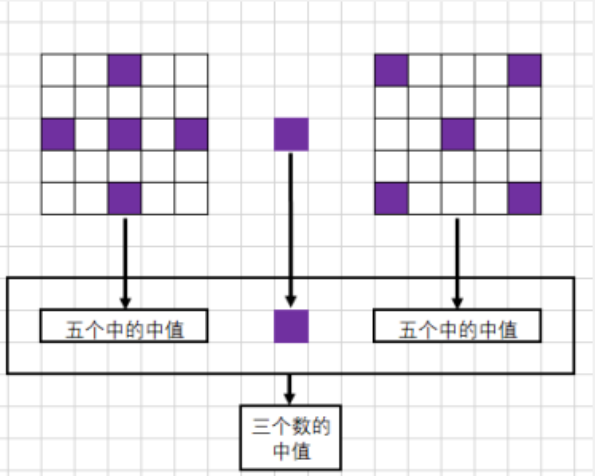
分别求两个窗口的中位值，结合当前窗口的中心点有3个候选值，再从中求出中位值作为结果



图解



扩展为 5*5

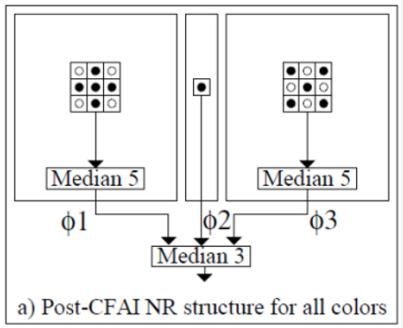


适用于 Bayer

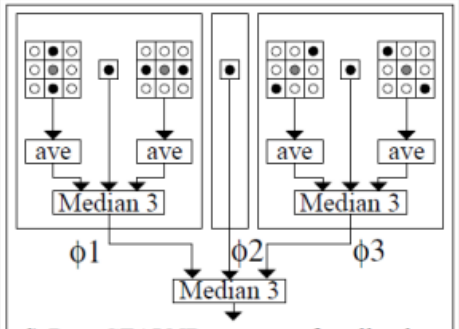
多级中值混合滤波=多级中值+均值

1. 求出水平和竖直方向相邻三个点的均值，再结合当前点，再求一个中位值
2. 求出对角线方向上的均值，然后结合当前点求出一个中位值
3. 两个中位值结合当前点，最后求一个中位值作为结果

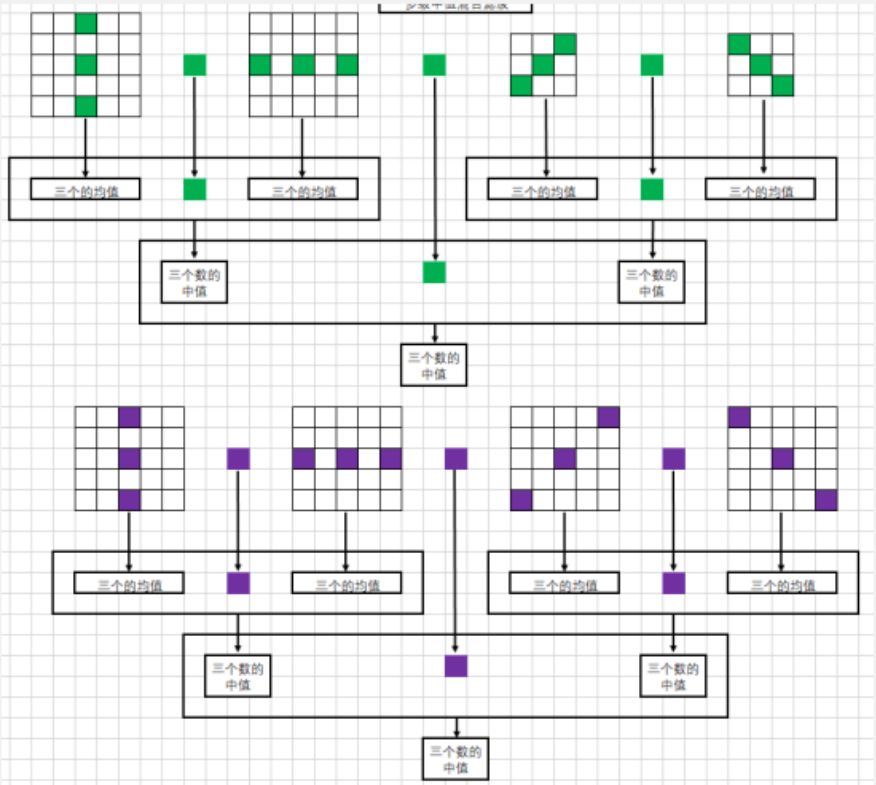
多级中值滤波 VS 多级中值混合滤波



a) Post-CFAI NR structure for all colors



d) Post-CFAI NR structure for all colors



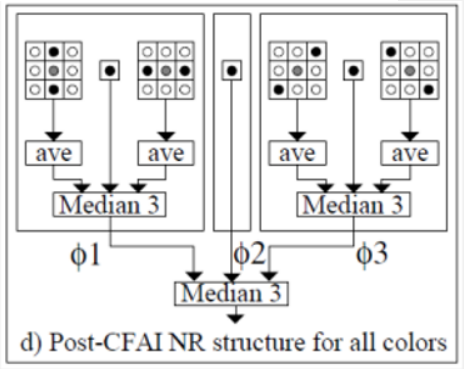
扩展为 5*5

适用于 Bayer

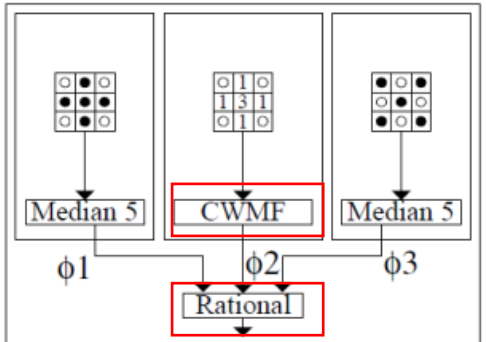
CWMF：加权中值滤波，和加权均值滤波一样，每个点给一个权重，在加权后的数据中取出中位值作为滤波后的值

Rational：
$$y(m,n) = \phi 2(m,n) + \frac{\phi 1(m,n) - 2 * \phi 2(m,n) + \phi 3(m,n)}{h + k(\phi 1(m,n) - \phi 3(m,n))}$$

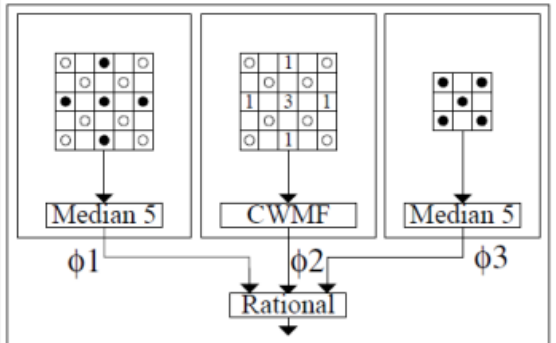
多级中值混合滤波　VS　多级中值有理混合滤波



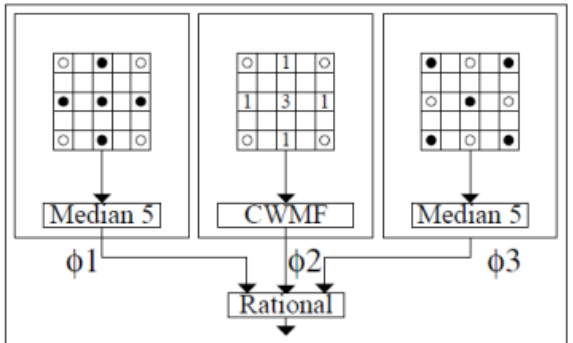
d) Post-CFAI NR structure for all colors



a) Post-CFAI NR window for all colors



b) Pre-CFAI NR window for green



c) Pre-CFAI NR window for red and blue

扩展为 5*5　　适用于 Bayer